

Lecture 2: Welcome to Class, Mr. SDF

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Intro

- Where do prices come from?
- If you ask your friend at an asset management company:

$$P_t = \sum_{i=1}^{\infty} \frac{\mathbb{E}[D_{t+i} | \mathcal{F}_t]}{(1 + r_t)^i}$$

- Key question: what's the right r_t ?
- Cochrane (2011): the central question of asset pricing is what drives r_t ;
- A theory about P_t is a theory about r_t , given cash-flow expectations;

Our agenda

- Let's do this in proper EPGE style: microfounded first principles;
- First: a two-period model to help us think about P_t, r_t ;
- Second: let's study a contingent claims economy to understand prices better;
- Third: do we actually need all this to price an asset? Functional Analysis to the rescue!

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Food for thought:

- Why would you ever hold an asset that pays off in the future?;
- Why do you care about returns?
- Why do you care about uncertainty?

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- Preferences captured by expected utility, with Bernoulli utility $u(c)$ and discount factor β ;
- She chooses how much to consume at t and $t + 1$;
- There is a single risky asset that pays a random payoff x_{t+1} ;
- At t : choose a savings (in ξ units of the asset) and a consumption plan;
- At $t + 1$: consume endowment + asset payoff;

The Consumer Problem

$$\max_{c_t, c_{t+1}, \xi} u(c_t) + \beta \mathbb{E}_t[u(c_{t+1})]$$

s.t.

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The FOC are:

$$u'(c_t) \cdot P_t = \beta \mathbb{E}_t[u'(c_{t+1})x_{t+1}] = \beta \mathbb{E}_t[u'(e_{t+1} + x_{t+1}\xi)x_{t+1}]$$

↓

$$P_t = \mathbb{E}_t \left[\beta \frac{u'(c_{t+1})}{u'(c_t)} \cdot x_{t+1} \right]$$

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- Let $m_{t+1} \equiv \beta \frac{u'(c_{t+1})}{u'(c_t)}$. We call this guy the Stochastic Discount Factor (SDF);
- This notation implies the most important equation in Asset Pricing:

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This class is about m_{t+1} . Anything else is a sideshow.

Questions?

Pricing Returns

Another Version of the Fundamental Equation

- Define the gross return on the asset as $R_{t+1} \equiv \frac{x_{t+1}}{P_t}$;
- Example in the of a stock: $x_{t+1} = D_{t+1} + P_{t+1}$;
- Rearranging the fundamental equation:

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- Now consider an asset with a *certain* return $R_f \implies$ a “risk-free” asset;
- It also holds that $\mathbb{E}[m_{t+1}R_t^f] = 1$;
- For any asset, $R_{t+1} - R_t^f$ is its *risk premium*;
- Rearranging the fundamental equation:

$$\mathbb{E}[m_{t+1}(R_{t+1} - R_t^f)] = 0$$

- From $\mathbb{E}[m_{t+1}R_t^f] = 1$, we have:

$$R_t^f = \frac{1}{\mathbb{E}_t[m_{t+1}]}$$

- In *any economy*, the risk-free rate is tightly connected to $\mathbb{E}_t[m_{t+1}]$;
- Importance for the literature: models for the risk-free interest rates are super important!

The Risk-Free Rate and the SDF

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- In *any economy*, the risk-free rate is tightly connected to $\mathbb{E}_t[m_{t+1}]$;
- Importance for the literature: models for the risk-free interest rates are super important!
- Challenge: what happens to the price P_t of the single risky asset if our consumer is risk-neutral?
- Tip: what would happen to his Bernoulli utility function?

When Do Assets Have High Returns?

- Recall: $\mathbb{E}_t[m_{t+1}(R_{t+1} - R_t^f)] = 0$ for any asset;
- Rearranging:

$$\mathbb{E}_t[R_{t+1} - R_t^f] = -\frac{\text{Cov}_t(m_{t+1}, R_{t+1})}{\mathbb{E}_t[m_{t+1}]} = -\text{Cov}_t(m_{t+1}, R_{t+1}) \cdot R_t^f$$

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- The most important object here: $\text{Cov}_t(m_{t+1}, R_{t+1})$;
- Recall: $m_{t+1} = \beta \frac{u'(c_{t+1})}{u'(c_t)}$;
- When is m_{t+1} high? When c_{t+1} is low (why?);

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- When is m_{t+1} high? When c_{t+1} is low (why?);
- If $\text{Cov}_t(m_{t+1}, R_{t+1}) < 0 \implies \mathbb{E}[R_{t+1} - R_t^f] > 0$. Why? Can you give an example?
- If $\text{Cov}_t(m_{t+1}, R_{t+1}) > 0 \implies \mathbb{E}[R_{t+1} - R_t^f] < 0$. Why? Can you give an example?

When Do Assets Have High Prices?

- Recall: $P_t = \mathbb{E}_t[m_{t+1}x_{t+1}]$ for any asset;
- Rearranging:

$$P_t = \frac{\mathbb{E}_t[x_{t+1}]}{R_t^f} + \text{Cov}_t(m_{t+1}, x_{t+1})$$

- Given cash-flow expectations, assets have high prices when they have lower risk premium;
- This happens when $\text{Cov}_t(m_{t+1}, x_{t+1}) > 0$;
- How is this related to what your friend at the asset management company does?

A Quick Look Into Infinite Periods

- Assume that the asset pays a stream of dividends d_t for $t = 0, 1, 2, \dots$;
- The consumer seeks to maximize $\mathbb{E}_t \left[\sum_{j=0}^{\infty} u(c_{t+j}) \right]$;
- The payoff at $t + 1$ is now $x_{t+1} = d_{t+1} + P_{t+1}$;
- Your job: show at home that

$$P_t = \mathbb{E}_t [m_{t+1} (P_{t+1} + d_{t+1})] = \mathbb{E}_t \left[\sum_{j=1}^{\infty} \left(\underbrace{\prod_{k=1}^j m_{t+k}}_{m_{t,t+j}} \right) d_{t+j} \right] = \mathbb{E}_t \left[\sum_{j=1}^{\infty} m_{t,t+j} d_{t+j} \right]$$

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- Exercise: show that, under a certain limiting condition, we have

$$P_t = \sum_{j=1}^{\infty} \frac{\mathbb{E}[d_{t+j}]}{R_{t,t+j}^f} + \sum_{j=1}^{\infty} \text{Cov}(d_{t+j}, m_{t,t+j})$$

Real Returns vs Nominal Returns

- The basic equation $\mathbb{E}[m_{t+1}R_{t+1} = 1]$ holds for both real and nominal returns;
- The main difference is the definition of m_{t+1} ;
- There is a *real* SDF (denominated in goods) and a *nominal* SDF (denominated in dollars);
- They differ due to inflation!
- If you assumed that P_t and x_{t+1} were nominal variables, you can write

$$\frac{P_t}{\Pi_t} = \mathbb{E}_t \left[m_{t+1} \frac{x_{t+1}}{\Pi_{t+1}} \right] \implies P_t = \mathbb{E}_t \left[m_{t+1} \frac{\Pi_t}{\Pi_{t+1}} x_{t+1} \right]$$

where Π_t is the price level and $m_{t+1} \frac{\Pi_t}{\Pi_{t+1}}$ is the nominal SDF;

Questions?

A Contingent Claims Economy

- We don't actually need utility functions to have an SDF;
- We don't need an equilibrium notion either;
- Next lecture: what are the minimal conditions for an SDF to exist?
- For now: what is the SDF with contingent claims?

- There are only two dates S states of nature;
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- Each state s occurs with probability $\pi_s > 0$;
- There are S Arrow-Debreu securities, or contingent claims;
- Security j pays 1 unit of account if state j occurs, 0 otherwise;
- Let q_j be the price of security j ;
- We assume these prices exist and they clear markets;
- Crucial assumption: the Law of One Price holds;

Pricing Any Asset

- Let's price an arbitrary asset that pays x_s units of account in state s ;
- By the Law of One Price, we can replicate this asset with Arrow-Debreu securities;
- The price of the asset is:

$$P = \sum_{s=1}^S q_s x_s$$

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- We can rewrite this as:

$$P = \sum_{s=1}^S \underbrace{\frac{q_s}{\pi_s}}_{\equiv m_s} \cdot \pi_s \cdot x_s = \sum_{s=1}^S \pi_s \cdot m_s \cdot x_s = \mathbb{E}[mx]$$

- We have found an SDF: $m_s = \frac{q_s}{\pi_s}$;
- Exercise: show at home that this SDF is unique, given prices q_s and probabilities π_s .

A Synthetic Risk-Free Asset

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- What's the return of this asset?

$$R^* \equiv \frac{1}{P^*} = \frac{1}{\mathbb{E}[m]} > 0$$

- This asset is risk-free by construction! So let's write $R^* = R^f$;

- Consider another probability distribution $\{\pi_1^*, \dots, \pi_S^*\}$ such that:

$$\pi_s^* \equiv R^f \cdot m_s \cdot \pi_s = R^f q_s = \frac{q_s}{\sum_{j=1}^S q_j}$$

- Notice that $\pi_s^* > 0$ and $\sum_{s=1}^S \pi_s^* = 1$;

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- Notice that $\pi_s^* > 0$ and $\sum_{s=1}^S \pi_s^* = 1$;
- Importantly, this probability measure prices assets *as if* everyone were risk-neutral:

$$P = \sum_{s=1}^S q_s x_s = \sum_{s=1}^S \pi_s^* \cdot \frac{x_s}{R^f} = \mathbb{E}^* \left[\frac{x}{R^f} \right]$$

where $\mathbb{E}^*$ is the expectation under the risk-neutral measure;

Intuition for the Risk-Neutral Measure

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- The risk-neutral measure shifts mass from good states to bad states;
- Pay a lot of attention to states that either very likely (high π_s) or really bad (high m_s);

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- Exercise: show that $q_s = \beta \pi_s \frac{u'(c_{t+1}(s))}{u'(c_t)}$;
- Using our definitions: $m(s) = \beta \frac{u'(c_{t+1}(s))}{u'(c_t)} \implies \frac{m(s_1)}{m(s_2)} = \frac{u'(c_{t+1}(s_1))}{u'(c_{t+1}(s_2))}$. Intuition?

Questions?

(C): Chapters 1 and 3;

Cochrane, John H. 2011. “Presidential Address: Discount Rates.” *The Journal of Finance* 66 (4): 1047–1108. <https://doi.org/10.1111/j.1540-6261.2011.01671.x>.